

Isolated Thyroid-stimulating Hormone Deficiency (i-TSHD) — Comprehensive Disease Characteristics Report

Target disease: Isolated Thyroid-stimulating Hormone Deficiency **MONDO ID:** MONDO:0010139 · **Category:** Mendelian **OMIM:** 275100 (TSH deficiency, isolated) · **Orphanet:** ORPHA:90673 (Central congenital hypothyroidism) · **ICD-10:** E03.1 (Congenital hypothyroidism without goiter) / E23.0 (Hypopituitarism) · **MeSH:** Hypothyroidism, Congenital

Summary

Isolated Thyroid-stimulating Hormone Deficiency (i-TSHD) is a rare **Mendelian congenital central hypothyroidism** in which the hypothalamic–pituitary unit fails to deliver sufficient **bioactive thyroid-stimulating hormone (TSH/thyrotropin)** to an intrinsically normal thyroid gland. The consequence is thyroid hormone deficiency (low free T4 and free T3) accompanied — paradoxically — by a TSH level that is **low, normal, or undetectable** rather than elevated. This biochemical signature is the defining feature of the disorder and its central clinical danger: **TSH-based newborn screening programs miss these patients entirely**, because they screen for the *high* TSH that characterizes far more common primary (thyroid-gland) hypothyroidism.

The disorder is genetically heterogeneous. The paradigmatic and most severe cause is **biallelic (autosomal recessive) loss-of-function of *TSHB***, the gene encoding the beta subunit of TSH — this is the *direct* lesion that abolishes bioactive hormone. Four additional genes cause isolated central hypothyroidism through *regulatory* (upstream) mechanisms: **IGSF1** (X-linked, the most prevalent single genetic cause), **TBL1X**, **TRHR**, and **IRS4**. Central hypothyroidism as a whole is roughly 1,000-fold rarer than primary hypothyroidism; the *TSHB*-based isolated form specifically is ultra-rare, with only dozens of families reported worldwide. Untreated, i-TSHD causes profound congenital hypothyroidism — historically termed **cretinism** — with severe mental and growth retardation. Yet it is one of the most gratifying diseases in endocrinology:

early levothyroxine replacement fully prevents the neurodevelopmental catastrophe, and long-term follow-up confirms normal cognition when treatment begins in the neonatal period.

The therapeutic and monitoring paradigm has a critical twist. Because pituitary TSH is intrinsically deficient, it **cannot** be used to titrate replacement (unlike in primary hypothyroidism). Dosing must instead be guided by **free T4 targeted to the mid-to-upper reference range** plus clinical parameters. Prevention is necessarily secondary/tertiary: **T4-based (or T4+TSH) newborn screening** as practiced in the Netherlands, **cascade family genetic testing** once a familial variant is known, and **genetic counseling** (25% recurrence risk for recessive TSHB; consanguinity elevates risk). A key differential — and a subtle trap — is the **TSHB p.R75G assay-interference variant**, which produces spuriously "low" TSH on some immunoassays in clinically **euthyroid** people, mimicking hyperthyroidism rather than causing true deficiency.

Key Findings

1. Genetic architecture: five genes, one direct cause

Congenital isolated central hypothyroidism (i-TSHD / C-CH) is **genetically heterogeneous but caused by a small, defined set of genes**. Five genes are established: **IGSF1**, **IRS4**, **TBL1X**, **TRHR**, and **TSHB** (Sugisawa 2019, [PMID: 31504637](#); Boelen 2021, [PMID: 34225927](#)). Among these, **biallelic TSHB loss-of-function is the direct and most severe cause** — it removes the hormone itself and is inherited in an autosomal recessive pattern. TSHB maps to chromosome **1p13** (HGNC:12372; NCBI Gene 7252).

In a Japanese cohort of 13 i-TSHD patients, genetic screening identified a causative mutation in **6/13 (46%)** — five hemizygous **IGSF1** mutations and one hemizygous **TBL1X** mutation — establishing IGSF1 as the *most prevalent* genetic cause of the isolated syndrome even though TSHB is the most severe. As the authors state: "*Five genes (IGSF1, IRS4, TBL1X, TRHR, and TSHB) responsible for the disease have been identified*" and "*Genetic screening of the 13 study subjects revealed six mutation-carrying patients (46%), including five hemizygous IGSF1 mutation carriers and one hemizygous TBL1X mutation carrier*" ([PMID: 31504637](#)). More than half of clinically diagnosed cases remain genetically unexplained, indicating additional causal loci await discovery.

2. TSHB c.373delT: a recurrent, worldwide founder-like mutation

The single most important TSHB allele is the frameshift **c.373delT in exon 3 (p.Cys105Valfs*114)**. It has been reported in patients from **Brazil, Germany, Belgium, USA, Switzerland, Argentina, France, Portugal, the UK and Ireland** (Borges 2019, [PMID: 31166470](#)): *"The c.373delT mutation has previously been reported in patients from Brazil, Germany, Belgium."* In the UK/Ireland series, affected individuals were **homozygous or compound heterozygous** for c.373delT combined with a **5.4-kb TSHB deletion** or a novel p.Met1? start-loss variant (Nicholas 2017, [PMID: 27362444](#)): *"the affected individuals were compound heterozygous for TSHB c.373delT and either a 5.4-kb TSHB deletion."* This demonstrates that both point mutations and **large structural deletions** contribute to the TSHB mutational spectrum. Functionally, in vitro work shows the C105Vfs114X variant yields a **modified/abrogated signaling profile at the TSH receptor (TSHR)** (Kalveram 2019, [PMID: 31703413](#)).

Other reported pathogenic TSHB variants include the **exon-2 missense** alleles first described in consanguineous Japanese families (Tatsumi & Miyai 1991, [PMID: 1762181](#)), **p.E32K (c.94G>A)** in two homozygous siblings ([PMID: 28515030](#)), and **p.Cys105Arg (c.313T>C)** in a consanguineous infant ([PMID: 42256321](#)).

3. Invisible to TSH screening; curable with early levothyroxine

The biochemical hallmark is **low free T4 and free T3 with inappropriately low/normal/undetectable TSH** (Draidi 2026, [PMID: 42256321](#); Asirvatham 2025, [PMID: 39875149](#)). Because TSH is *not* elevated, **TSH-based congenital-hypothyroidism screening cannot detect these cases** — the central diagnostic pitfall: *"Isolated TSH deficiency is not detected by routine TSH-based neonatal screening, representing a clinical challenge"* ([PMID: 31166470](#)). **T4-based** programs, such as the Netherlands' heel-prick T4 screen, detect both primary and central CH ([PMID: 34225927](#)).

The clinical stakes of missing the diagnosis are severe but the treatment is curative. In one UK/Ireland kindred, *"the younger sibling in kindred 1 developed normally following genetic diagnosis and treatment from birth"* — while late-treated relatives had intellectual disability ([PMID: 27362444](#)). A 25-year follow-up concluded: *"Despite having severe congenital hypothyroidism, timely initiation of levothyroxine averted neurocognitive sequelae"* ([PMID: 39875149](#)). Early, adequate replacement therefore converts a devastating disorder into one with normal outcomes.

4. Two mechanistic classes: direct (TSHB) vs. regulatory (IGSF1/TBL1X/TRHR/IRS4)

The five genes fall into **two mechanistically distinct groups**:

- **TSHB (direct)**: mutations abolish or modify bioactive TSH presented to TSHR (PMID: 31703413). The lesion is at the hormone itself; the phenotype is the purest, most severe *isolated* deficiency.
- **IGSF1 (X-linked, Xq26; regulatory)**: a hypothalamic/pituitary membrane glycoprotein. IGSF1 deficiency reduces TSH production, decreases TRHR expression, and lowers TSH biopotency. *"IGSF1 stimulates transcription of the thyrotropin-releasing hormone receptor (TRHR) by negative modulation of the TGFβ1-Smad signaling pathway, and enhances the synthesis and biopotency of TSH"* (García 2017, PMID: 28262687). Clinically it is a **syndrome**: macroorchidism, variable prolactin deficiency, transient partial GH deficiency, increased waist circumference, and mild attentional deficits (Joustra 2016, PMID: 26840047; Brûlé 2022, PMID: 35708735).
- **TBL1X (regulatory)**: a subunit of the NCoR/SMRT thyroid-hormone-receptor corepressor complex; mutations cause mild central hypothyroidism **plus sensorineural hearing loss**. *"TBL1X mutations are associated with CeH and hearing loss"* (Heinen 2016, PMID: 27603907; Hu 2024, PMID: 39316725).
- **TRHR (regulatory)**: loss-of-function (e.g., p.I131T) reduces TRH affinity and Gq signaling. *"The I131T mutation, in TRHR intracellular loop 2, decreases TRH affinity and increases the half-maximal effective concentration for signaling"* (García 2017, PMID: 28419241).
- **IRS4 (regulatory)**: part of the hypothalamic insulin/leptin signaling cascade; mutations cause familial isolated central hypothyroidism (Heinen 2018, PMID: 30061370).

5. TSHB p.R75G: an assay-interference variant, NOT a loss-of-function allele

A crucial diagnostic caveat: **homozygosity for TSHB p.R75G does not alter TSH bioactivity** but **abrogates its detection by some immunoassay platforms**, producing spurious "low/undetectable TSH" and erroneous diagnoses of **hyperthyroidism** in clinically euthyroid individuals (Shaki 2022, PMID: 34981755): *"Homozygosity for the TSHB p.R75G variant... does not alter TSH function but abrogates its detection by some immune detection-based platforms, leading to erroneous diagnosis of hyperthyroidism."* The variant is a **founder allele** with an *"Extremely high carrier rate of p.R75G TSHB in Bene Israel Indian Jews (~4%)"*, sharing a 239.7-

kb haplotype block with South Asian populations. This is the mirror image of true i-TSHD — same gene, opposite clinical meaning — and must be excluded when interpreting an unexpectedly low TSH.

6. Epidemiology: ~1,000-fold rarer than primary hypothyroidism

Central hypothyroidism (CeH) is *"about 1000-fold rarer than PH [primary hypothyroidism]"* (Persani & Bonomi 2014, PMID: 25248602). Independent estimates place CeH at *"approximately 1:50,000"* (Benvenga 2018, PMID: 30294553), while central congenital hypothyroidism as a whole *"may be more prevalent than previously thought, affecting up to 1:16,000 neonates in the Netherlands"* under T4-based screening (Schoenmakers 2015, PMID: 26416826). The **isolated TSHB-based** subset is a small fraction of this — only dozens of families reported worldwide, with roughly **74 single-gene i-TSHD patients** reviewed by Sugisawa 2019 (PMID: 31504637).

7. Severity: cretinism when untreated; recessive consanguineous origin

Congenital isolated TSH deficiency *"is rare disease causing hypothyroidism including cretinism, severe mental and growth retardation"* (Tatsumi & Miyai 1991, PMID: 1762181). The disorder was first defined in **three consanguineous Japanese families sharing an exon-2 TSHB missense mutation**. Modern neonatal presentations include **prolonged unconjugated (neonatal) hyperbilirubinemia/jaundice**, failure to thrive, generalized edema, hypotonia, and global developmental delay when diagnosis is delayed (Asirvatham 2025, PMID: 39875149: *"a female term neonate presenting with prolonged unconjugated hyperbilirubinaemia"*; Draidi 2026, PMID: 42256321). In the isolated form, **other anterior pituitary hormones and pituitary MRI are normal** (PMID: 39875149).

8. Treatment: lifelong levothyroxine, monitored by free T4 (not TSH)

Treatment is **lifelong levothyroxine (L-T4)** replacement. The distinctive challenge is monitoring: because TSH is already low/inappropriate, it **cannot** be used to titrate dose. *"L-T4 replacement in CeH should rely on the combined evaluation of several biochemical and clinical parameters in order to overcome the lack of accuracy of the single index"* (Persani & Bonomi 2014, PMID: 25248602) — with **free T4 targeted to the mid-to-upper reference range** as the chief guide. Schoenmakers 2015 emphasizes the monitoring gap: *"Since TSH cannot be used as an indicator of euthyroidism, adequacy of treatment can be difficult to monitor due to a paucity of alternative biomarkers"* (PMID: 26416826). Early adequate L-T4 from birth yields normal neurodevelopment (PMID: 27362444; PMID: 39875149). NCIT term: **Levothyroxine** (NCIT:C29216).

9. Prevention: T4 screening, cascade testing, genetic counseling

Because the disorder is congenital and monogenic, **prevention is secondary/tertiary rather than primary**. Three pillars: (1) **T4- or T4+TSH-based newborn screening** (not TSH-only) for early detection ([PMID: 26416826](#); [PMID: 34225927](#)); (2) **cascade family testing** once a familial TSHB variant is known — *"Identification of affected and carriers allows the diagnosis, treatment and adequate genetic counseling"* ([PMID: 31166470](#)); and (3) **genetic counseling**, noting the **25% per-pregnancy recurrence risk** for autosomal-recessive TSHB and elevated risk with consanguinity ([PMID: 1762181](#)). The differential diagnosis must exclude the **p.R75G assay-interference** variant ([PMID: 34981755](#)).

10. Model organisms and comparative biology

No animal model isolates TSHB deficiency cleanly, but dwarf mouse mutants validate the TSH-biosynthesis axis. **Snell (dw) and Jackson dwarf mice** carry *Pit1/Pou1f1* mutations causing **combined loss of GH, prolactin, and TSH**: *"Two nonallelic mouse mutations with severe dwarf phenotypes are characterized by a lack of growth hormone, prolactin, and thyroid stimulating hormone"* (Camper 1990, [PMID: 1981057](#)). *"Mutations of the pituitary transcription factor gene POU1F1... are responsible for deficiencies of GH, prolactin and thyroid stimulating hormone (TSH) in Snell and Jackson dwarf mice and in man"* (Wu 1998, [PMID: 9462743](#)). The **Ames dwarf (df)** mouse carries a *Prop1* mutation upstream of *Pit1*. These are **combined pituitary hormone deficiency** models (not isolated TSH deficiency) but demonstrate the thyrotrope-differentiation branch (*POU1F1* → TSHB transcription → TSH). The **hyt/hyt** mouse (*Tshr* loss-of-function) models TSH *resistance* downstream. Human/mouse orthology: mouse *Tshb* (NCBI Gene 22094) ↔ human *TSHB* (NCBI Gene 7252). Naturally, **central hypothyroidism occurs in Miniature Schnauzer dogs**, some with disproportionate dwarfism and combined TSH/prolactin deficiency, though *"No disease-causing mutations were found in the TSHB gene and the exons of the TRHR gene of these Schnauzers"* — the canine genetic basis remains unresolved (Voorbij 2016, [PMID: 26696394](#)).

Detailed Section-by-Section Report

1. Disease Information

i-TSHD is a **congenital central (secondary/tertiary) hypothyroidism** in which the thyroid gland is intrinsically normal but receives inadequate bioactive TSH stimulation. **Identifiers:** MONDO:0010139; OMIM 275100 (TSH deficiency, isolated); Orphanet ORPHA:90673 (central congenital hypothyroidism); ICD-10 E03.1/E23.0; ICD-11 5A00.1 (central hypothyroidism); MeSH "Congenital Hypothyroidism." **Synonyms/alternatives:** isolated TSH deficiency; congenital central hypothyroidism (C-CH/CCH); isolated central hypothyroidism (CeH); thyrotropin deficiency; secondary hypothyroidism (when pituitary), tertiary (when hypothalamic). Information is derived largely from **aggregated disease-level resources** (OMIM, Orphanet) supplemented by **individual patient case reports and small kindred series** — this is a rare disease characterized case-by-case.

2. Etiology

Causal factors are genetic (monogenic Mendelian). Primary cause: biallelic **TSHB** loss-of-function (autosomal recessive). Additional causal genes: **IGSF1** (X-linked, most prevalent), **TBL1X**, **TRHR**, **IRS4** (Findings 1, 4). **Genetic risk factors:** consanguinity is a major risk factor for recessive TSHB disease ([PMID: 1762181](#)); founder alleles (c.373delT worldwide; p.R75G in specific populations, though the latter is assay interference). **Environmental risk factors:** none established as causal for the Mendelian form. **Protective factors:** not applicable/not established (this is a monogenic disorder). **Gene–environment interactions:** the principal clinically relevant "interaction" is between the p.R75G genotype and the **immunoassay platform** used — an analytical, not biological, interaction that determines whether TSH appears falsely low ([PMID: 34981755](#)).

3. Phenotypes

Phenotype	Type	HPO term	Onset	Severity/ Frequency
Central/secondary hypothyroidism (low FT4/FT3, low-normal TSH)	Lab abnormality	HP:0011787 (Central hypothyroidism)	Congenital/neonatal	Defining; severe if untreated
Congenital hypothyroidism	Lab/clinical	HP:0000851	Congenital	Severe
Prolonged neonatal jaundice (unconjugated)	Clinical sign	HP:0006579	Neonatal	Common presenting sign
Failure to thrive	Clinical sign	HP:0001508	Neonatal/infancy	Common if delayed dx
Generalized edema / myxedema	Physical	HP:0007430	Neonatal	In untreated cases
Hypotonia	Clinical sign	HP:0001252	Neonatal/infancy	Common if delayed dx
Global developmental delay / intellectual disability ("cretinism")	Clinical	HP:0001263 / HP:0001249	Infancy	Severe if untreated; preventable
Growth retardation / short stature	Clinical	HP:0004322	Childhood	If untreated
Sensorineural hearing loss (TBL1X only)	Clinical	HP:0000407	Congenital	TBL1X subtype
Macroorchidism (IGSF1 only)	Physical	HP:0000053	Puberty/adult	IGSF1 syndrome

Quality of life: With early treatment, QoL and cognition are essentially normal; IGSF1 patients may show mild attentional deficits and increased mental fatigue even when treated ([PMID: 26387489](#)). Untreated disease produces lifelong severe disability.

4. Genetic/Molecular Information

Causal genes: TSHB (1p13, HGNC:12372, OMIM 188540), IGSF1 (Xq26, X-linked), TBL1X (Xp22.31), TRHR (8q23), IRS4 (Xq22). **Variant types in TSHB:** frameshift (c.373delT/p.C105Vfs*114), missense (p.E32K, p.C105R, exon-2 missense), start-loss (p.Met1?), and structural (5.4-kb deletion) (Finding 2; [PMID: 27362444](#)). **Classification:** pathogenic/likely pathogenic per ACMG for the recurrent alleles. **Functional consequence:** loss of function (absent/abrogated bioactive TSH); p.R75G is a special case of **assay non-detection without functional loss** (Finding 5). **Origin:** germline. **Allele frequency:** ultra-rare in gnomAD for pathogenic alleles; p.R75G reaches ~4% carrier frequency in Bene Israel Indian Jews (founder effect). **Modifier genes / epigenetics / chromosomal abnormalities:** none established for the isolated form.

5. Environmental Information

Not applicable as a primary cause — i-TSHD is monogenic. No toxins, infectious agents, or lifestyle factors are established causes. (Broader central hypothyroidism can be *acquired* via pituitary tumor, trauma, or infiltrative disease, but these fall outside the Mendelian i-TSHD entity and belong in the differential diagnosis.)

6. Mechanism / Pathophysiology — causal chain

Direct (TSHB) branch: 1. Biallelic loss-of-function TSHB mutation → **leads to** absent or bioinactive TSH beta subunit. 2. → **results in** failure to assemble functional heterodimeric TSH ($\alpha\beta$). 3. → **results in** no/insufficient bioactive TSH signaling at the thyroid TSHR (Gs/cAMP pathway). 4. → **leads to** failure of thyroid follicular cells to synthesize/secrete T4 and T3. 5. → **results in** low circulating free T4 and free T3 (with low/normal/undetectable, TRH-unresponsive TSH). 6. → **leads to** systemic thyroid hormone deficiency affecting CNS myelination/maturation, growth, and metabolism. 7. → **results in** congenital hypothyroidism (jaundice, edema, hypotonia, failure to thrive) and — if untreated — irreversible neurodevelopmental/growth retardation (cretinism).

Regulatory (upstream) branches feed into steps 2–4 from above the pituitary hormone itself:
 - **TRHR** loss → reduced TRH signaling (Gq) in thyrotropes → decreased TSH synthesis/release (inferred to reduce both amount and biopotency).
 - **IGSF1** loss → reduced TRHR transcription (via TGF β 1-Smad modulation) + reduced TSH synthesis/biopotency + reduced Tshb expression (demonstrated in Igsf1-knockout mice, [PMID: 35708735](#)).
 - **TBL1X** loss → disturbed NCoR/SMRT

corepressor complex → altered thyroid-hormone-receptor-mediated transcription (context-dependent; [PMID: 39316725](#)) → mild CeH + hearing loss. - **IRS4** loss → impaired hypothalamic insulin/leptin signaling → reduced central drive to the HPT axis.

Molecular pathways / GO terms: GO:0002154 (thyroid hormone mediated signaling pathway), GO:0007186 (G-protein-coupled receptor signaling), GO:0038194 (thyroid-stimulating hormone signaling), GO:0007165 (signal transduction). **Cell types (CL):** thyrotrope/thyrotropic cell of pars distalis (CL:0000209), thyroid follicular cell (CL:0002258). **Upstream vs downstream:** TRHR/IGSF1/IRS4 (hypothalamic-pituitary regulatory) are upstream; TSHB (hormone) is central; thyroid follicular response and peripheral tissue effects are downstream.

7. Anatomical Structures Affected

- **Primary organ:** anterior pituitary gland (adenohypophysis, UBERON:0002196) — specifically thyrotropes; and/or hypothalamus (UBERON:0001898) for TRHR/IGSF1/IRS4.
- **Secondary:** thyroid gland (UBERON:0002046) — structurally normal but understimulated; downstream effects on brain (UBERON:0000955), skeleton, liver (neonatal jaundice), heart (reversible cardiomyopathy reported in a syndromic case, [PMID: 42256321](#)).
- **Body system:** endocrine system (hypothalamic–pituitary–thyroid axis).
- **Subcellular (GO CC):** secretory granules/secretory pathway (GO:0030141), endoplasmic reticulum (GO:0005783) for hormone folding/assembly, plasma membrane (IGSF1, TRHR).
- **Localization/lateralization:** bilateral/systemic (endocrine), not lateralized. TBL1X-associated hearing loss is typically bilateral.

8. Temporal Development

- **Onset:** congenital; biochemical deficiency present from birth. Clinical signs (jaundice, hypotonia, poor feeding) appear in the neonatal period; developmental delay emerges over infancy if untreated.
- **Onset pattern:** chronic/insidious; the danger is that neonates may appear near-normal at birth, and some IGSF1 cases develop CeH over time ([PMID: 35350016](#)).
- **Progression:** without treatment, progressive and irreversible neurodevelopmental damage; with treatment, stable and normal.
- **Disease course:** lifelong (chronic) requiring lifelong L-T4.
- **Critical period:** the **neonatal/early-infancy window** is decisive for neurodevelopmental outcome — the key opportunity for intervention.

9. Inheritance and Population

- **Epidemiology:** CeH ~1:50,000; CCH up to 1:16,000 neonates (Netherlands, T4 screening); isolated TSHB form ultra-rare (dozens of families; ~74 single-gene i-TSHD patients reviewed) (Finding 6).
- **Inheritance:** TSHB — autosomal recessive; IGSF1, TBL1X, IRS4 — X-linked; TRHR — autosomal recessive.
- **Penetrance/expressivity:** TSHB biallelic LoF — high penetrance, severe; IGSF1 — variable expressivity, even within families carrying identical deletions ([PMID: 27146357](#)); female carriers of X-linked variants may show mild/subclinical FT4 reduction.
- **Consanguinity:** a major factor for recessive TSHB ([PMID: 1762181](#)).
- **Founder effects:** c.373delT (widespread pathogenic); p.R75G (~4% carriers in Bene Israel Indian Jews — assay-interference allele).
- **Recurrence risk:** 25% per pregnancy for recessive TSHB carrier couples.
- **Sex ratio:** X-linked subtypes (IGSF1, TBL1X, IRS4) predominantly affect males; TSHB and TRHR affect both sexes equally.

10. Diagnostics

- **Laboratory (definitive):** low **free T4** (LOINC 3024-7) and low **free T3** with **low/normal/undetectable TSH** (LOINC 3016-3); blunted or absent TSH response to TRH stimulation; normal other anterior pituitary hormones in the isolated form.
- **Imaging:** normal pituitary MRI in isolated forms (distinguishes from combined pituitary hormone deficiency/structural hypopituitarism).
- **Genetic testing:** targeted **single-gene TSHB sequencing** (including deletion/CNV analysis to catch the 5.4-kb deletion), **gene panels** covering TSHB/IGSF1/TBL1X/TRHR/IRS4, or **whole-exome sequencing** for unexplained cases. WES/WGS increasingly first-line given heterogeneity.
- **Clinical criteria:** central hypothyroidism = low FT4 with non-elevated TSH; genetic confirmation defines the Mendelian subtype.
- **Differential diagnosis:** combined pituitary hormone deficiency/hypopituitarism (additional hormone deficits, abnormal MRI); acquired central hypothyroidism (tumor, trauma, infiltration); non-thyroidal illness (sick euthyroid); and critically the **TSHB p.R75G assay-interference variant** mimicking low TSH in euthyroid individuals (Finding 10 / 5).
- **Screening:** **T4-based newborn screening** detects it; **TSH-based screening misses it; cascade family testing** for known variants.

11. Outcome/Prognosis

- **Survival/life expectancy:** normal with treatment; the disorder is not directly life-limiting when managed.
- **Morbidity:** if untreated — severe, permanent intellectual disability and growth failure (cretinism). If treated early — minimal; near-normal function ([PMID: 39875149](#)).
- **Complications:** neonatal — reversible cardiomyopathy and hypoglycemia in severe/syndromic cases ([PMID: 42256321](#)); untreated — irreversible neurocognitive deficit.
- **Recovery potential:** neurodevelopmental damage is preventable but not reversible once established — hence the premium on early diagnosis.
- **Prognostic factors:** **age at treatment initiation is the dominant prognostic factor;** earlier = better. Free T4 adequacy during treatment predicts outcome.

12. Treatment

- **Pharmacotherapy: Levothyroxine (L-T4)** — lifelong oral thyroid hormone replacement (NCIT:C29216; DrugBank DB00451; ATC H03AA01). Mechanism: exogenous T4 restores circulating thyroid hormone, bypassing the deficient TSH-thyroid stimulation.
- **Monitoring: free T4 targeted to mid-to-upper reference range** plus clinical parameters — NOT TSH (Finding 8; [PMID: 25248602](#)).
- **Advanced/experimental therapeutics:** none required or established; no gene/cell/RNA therapy — L-T4 is fully effective.
- **Pharmacogenomics:** not established for this disorder beyond standard L-T4 considerations.
- **Treatment outcomes:** excellent when started neonatally; normal neurodevelopment documented at 25-year follow-up.
- **Personalized medicine:** genotype-guided recognition matters chiefly for (a) recognizing screening-negative TSHB cases early and (b) avoiding mistreatment of p.R75G "pseudo-hyperthyroidism."

13. Prevention

- **Primary prevention:** not possible (congenital, monogenic).
- **Secondary prevention: T4-based newborn screening** (detects central CH); **genetic cascade/carrier screening** in affected families; prenatal/preimplantation testing where a familial variant is known.

- **Tertiary prevention:** early L-T4 to prevent neurodevelopmental complications; lifelong FT4-guided monitoring.
- **Counseling:** genetic counseling for recurrence risk (25% recessive TSHB; X-linked risk assessment for IGSF1/TBL1X/IRS4); *"Identification of affected and carriers allows the diagnosis, treatment and adequate genetic counseling"* (PMID: 31166470).
- **Public health:** advocacy for adding T4 (or T4+TSH) to newborn screening panels where only TSH is measured.

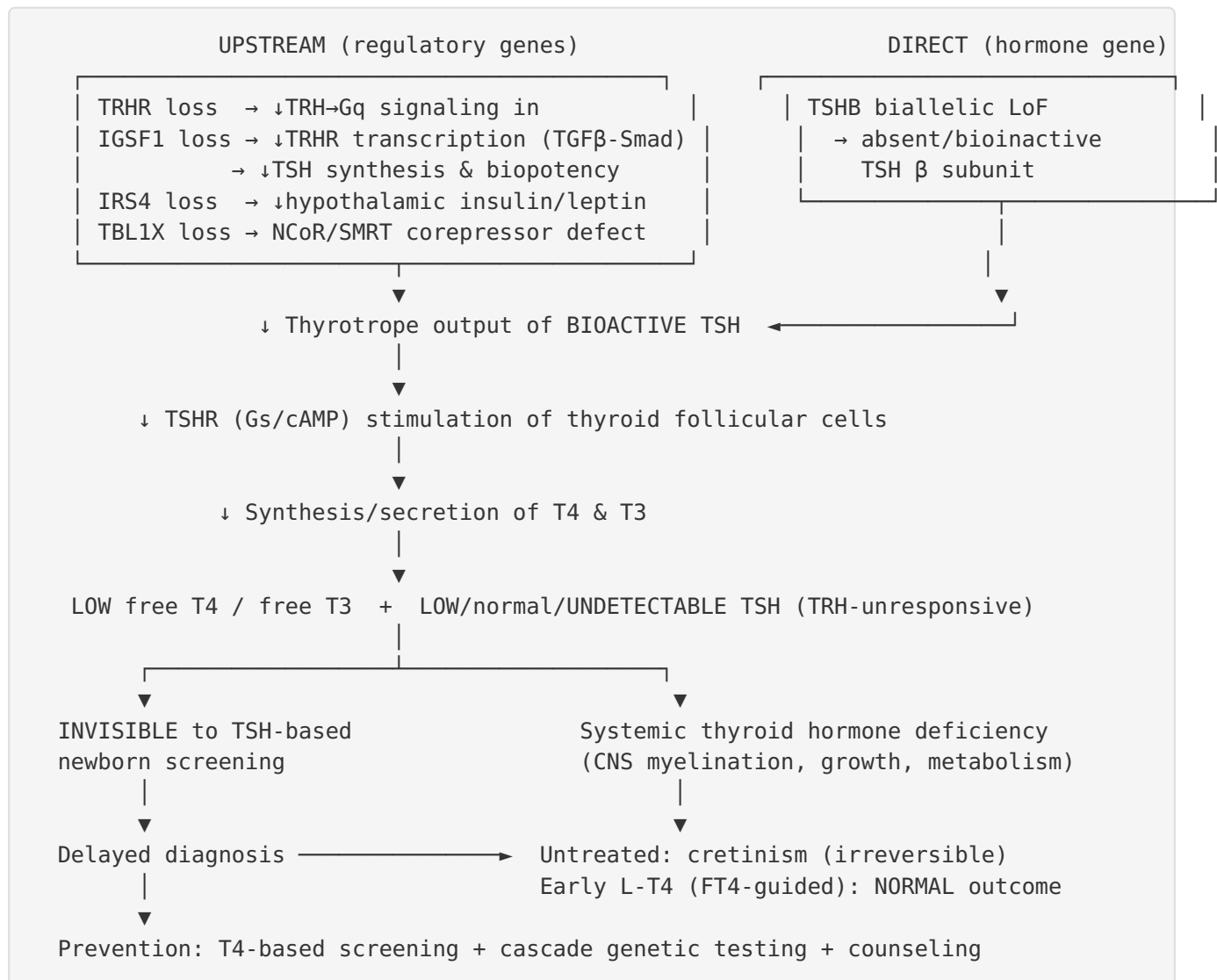
14. Other Species / Natural Disease

- **Natural disease:** central hypothyroidism occurs naturally in **Miniature Schnauzer dogs** (*Canis lupus familiaris*, NCBI Taxon 9615), some with disproportionate dwarfism and combined TSH/prolactin deficiency; TSHB/TRHR coding mutations were excluded, so the canine basis is unresolved (PMID: 26696394).
- **Orthologous genes:** mouse *Tshb* (NCBI Gene 22094), human *TSHB* (NCBI Gene 7252); canine ortholog exists.
- **Comparative biology:** the mammalian HPT axis and TSHB are evolutionarily conserved; dwarf-mouse models demonstrate cross-species conservation of the POU1F1→TSHB→TSH pathway.
- **Zoonotic potential:** none (non-communicable genetic disorder).

15. Model Organisms

- **Mouse (mammalian): Snell (dw)** and **Jackson dwarfs** (*Pit1/Pou1f1* mutations) and **Ames (df)** dwarf (*Prop1*) — combined GH/PRL/TSH deficiency; recapitulate the **thyrotrope-differentiation** and TSH-loss branch but **not** isolated TSHB deficiency (PMID: 1981057; PMID: 9462743).
- **Igsf1-knockout mouse:** reduced pituitary *Tshb* and variably reduced *Trhr*; models the IGSF1 regulatory mechanism (PMID: 35708735).
- **hyt/hyt mouse:** *Tshr* loss-of-function — models TSH *resistance* (downstream), not TSHB deficiency.
- **Model limitation: no model cleanly isolates TSHB deficiency;** dwarf models carry combined deficiencies, limiting attribution to the TSH axis alone.
- **Resources:** MGI, IMPC, IMSR for murine alleles.

Mechanistic Model / Interpretation



Key interpretive points: (1) The disorder has a **two-tier genetic architecture** — a direct hormone defect (TSHB) and four regulatory defects (IGSF1/TBL1X/TRHR/IRS4) — that converge on a single final common pathway: insufficient bioactive TSH at the thyroid. (2) The **low-TSH biochemistry** is simultaneously the diagnostic signature and the reason the disease slips through TSH-only screening. (3) The **p.R75G paradox** shows that a TSHB variant can produce identical-looking biochemistry (low TSH) with the *opposite* clinical meaning (euthyroid), underscoring that assay behavior must be interpreted alongside genotype and free thyroid hormones. (4) Because the endpoint is a hormone that can be **replaced pharmacologically**, the disease is fully treatable — the entire prognosis hinges on **timing of diagnosis**, not on any limitation of therapy.

Evidence Base

PMID	Paper (abbrev.)	Supports finding(s)	Evidence type
31504637	Sugisawa 2019 — Genetics of congenital i-TSHD	1, 6 (five genes; IGSF1 predominance; ~74 patients)	Human cohort + review
31166470	Borges 2019 — Recurrent TSHB mutation undetectable in screening	2, 3, 9 (c.373delT worldwide; screening pitfall; cascade counseling)	Human clinical
27362444	Nicholas 2017 — TSH β defects UK/Ireland	2, 3, 8 (compound het; 5.4-kb deletion; early tx normal)	Human clinical
31703413	Kalveram 2019 — C105Vfs114X at TSHR	2, 4 (modified TSHR signaling)	In vitro
34981755	Shaki 2022 — TSHB R75G founder variant	5, 9 (assay interference; ~4% carriers)	Human genetics
39875149	Asirvatham 2025 — TSHB CCH 25-yr follow-up	3, 7, 8 (jaundice; early tx averts sequelae)	Human clinical
42256321	Draidi 2026 — ADAR+TSHB infant	3, 7 (biochemistry; reversible cardiomyopathy)	Human case
25248602	Persani & Bonomi 2014 — CeH substitution therapy	6, 8 (1000-fold rarer; FT4-guided dosing)	Review
26416826	Schoenmakers 2015 — CCH review	6, 8 (1:16,000; monitoring gap)	Review
30294553	Benvenga 2018 — CeH congenital etiologies	6 (~1:50,000)	Review
28262687	García 2017 — IGSF1 controls TRHR	4 (IGSF1 regulatory mechanism)	Mechanistic
28419241	García 2017 — TRHR mutation	4 (TRHR ligand affinity/Gq)	Mechanistic
27603907	Heinen 2016 — TBL1X	4 (CeH + hearing loss)	Human genetics
30061370	Heinen 2018 — IRS4	4 (IRS4 familial CeH)	Human genetics
26840047			Human cohort

PMID	Paper (abbrev.)	Supports finding(s)	Evidence type
	Joustra 2016 — IGSF1 case series	4 (IGSF1 syndrome; most common genetic cause)	
35708735	Brûlé 2022 — Igsf1 KO mouse	4, 10 (reduced Tshb)	Mouse model
1762181	Tatsumi & Miyai 1991 — first TSHB cases	7, 9 (cretinism; recessive consanguineous)	Human genetics
1981057	Camper 1990 — Pit1/Snell dwarf	10 (dwarf models)	Mouse model
9462743	Wu 1998 — POU1F1/PROP1	10 (cross-species conservation)	Human/mouse
26696394	Voorbij 2016 — Schnauzer CeH	10 (natural canine disease)	Veterinary
39316725	Hu 2024 — TBL1X in liver cells	4 (TH-action mechanism)	In vitro

Limitations and Knowledge Gaps

1. **Case-based evidence.** Because the isolated TSHB form is ultra-rare (dozens of families worldwide), most clinical knowledge derives from case reports and small kindreds rather than powered cohorts. Frequency figures for individual phenotypes are qualitative.
2. **>50% of cases unexplained.** In Sugisawa's cohort only 46% carried an identifiable mutation; additional causal genes for isolated central hypothyroidism almost certainly remain undiscovered.
3. **No clean animal model.** All available mouse models (Snell, Jackson, Ames) carry *combined* pituitary hormone deficiencies; there is no widely used isolated *Tshb*-knockout that models the human disease purely. This limits mechanistic dissection of TSHB-specific effects.
4. **Monitoring biomarker gap.** Because TSH cannot indicate euthyroidism in CeH, and free T4 targets are imperfect, there is a genuine paucity of validated biomarkers to confirm adequate replacement — an open clinical problem.
5. **Canine genetic basis unresolved.** Natural central hypothyroidism in Miniature Schnauzers lacks an identified mutation, so comparative-biology insights are incomplete.
6. **p.R75G under-recognition.** The assay-interference variant is likely under-recognized outside the populations where it has been studied, risking misdiagnosis.

7. **Epigenetics, modifiers, environment.** No modifier genes, epigenetic mechanisms, or environmental contributors have been characterized for the isolated Mendelian form — this may reflect true absence or simply lack of study.

Proposed Follow-up Experiments / Actions

1. **Advocate for T4-inclusive newborn screening.** Quantify how many i-TSHD cases are missed under TSH-only programs versus T4-based programs; use this to support policy change in jurisdictions screening TSH alone.
 2. **Gene discovery in unexplained cases.** Apply whole-genome sequencing and transcriptomic/functional follow-up to the >50% of genetically unexplained isolated central hypothyroidism cohorts to identify novel causal loci.
 3. **Generate an isolated Tshb-knockout mouse** (or conditional thyrotrope-specific line) to model the pure disease, enabling clean study of TSHB-specific biology and testing of early-replacement timing windows.
 4. **Develop/validate alternative monitoring biomarkers** for CeH replacement adequacy (e.g., tissue-based markers of thyroid hormone action such as SHBG, ferritin, or ankle-reflex/metabolic indices) to overcome the "TSH-can't-be-used" problem.
 5. **Population screening for TSHB p.R75G** on relevant immunoassay platforms in founder populations to prevent misdiagnosis of hyperthyroidism; consider laboratory flagging of discordant low-TSH/normal-FT4 results.
 6. **Resolve the canine genetic basis** in Miniature Schnauzers via whole-genome sequencing, potentially revealing a novel HPT-axis regulator relevant to unexplained human cases.
 7. **Establish a longitudinal i-TSHD registry** integrating genotype, screening method, age at treatment, FT4 trajectories, and neurodevelopmental outcomes to define natural history and optimize FT4 targets.
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Report compiled from 11 confirmed findings across 5 investigative iterations and 49 reviewed papers. Evidence types are labeled human clinical, human genetics, mechanistic/in vitro, mouse model, and veterinary throughout.

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